

# Agent360: A Concealment-First Bilateral Negotiator for ANL 2026

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## The Goal

ANL score = **Advantage** + **Concealing**  
**Advantage** — close good deals above reserved value  
**Concealing** — hide true preferences from the opponent's learner  
Design principle: concealment first, then extract the deal

## Architecture

Bid stream only — no opponent class, no true utility



## Bidding Persona

Relative time  $t \rightarrow$



|                                                                                               |
|-----------------------------------------------------------------------------------------------|
| <b>Decoy</b><br>Wrong issue priorities · maximal mismatch · seat-asymmetric                   |
| <b>Transition</b><br>Mix decoy + truth · slower reveal when opening · early exit if conceding |
| <b>Closing</b><br>Blend our utility + theirs · mode caps · discount bait                      |

**Seat asymmetry:** opening seat keeps a longer decoy and stricter gates

## What We Rejected

**Reverse-psychology** (truth early) — hurts Concealing vs learners  
**Mid-game escape accept** — took mediocre / deceptive deals

## Opponent Model

**Smith frequency** — base preference estimate from their bids  
**Recency · time · issue blends** — trust late and diverse signals more  
**Trajectory** — concession slope · bait jumps · predicted path  
Five modes: mirror · learner · deceptive · conceding · unknown

## Mode Response

|           |                                    |
|-----------|------------------------------------|
| Conceding | Early decoy exit · chase agreement |
| Learner   | Full blend · no bait discount      |
| Deceptive | Discount Smith · reject bait       |
| Mirror    | Plain Smith · anti-mirror bids     |
| Unknown   | Conservative full blend            |

## Acceptance

Catastrophe floor near deadline  
Mode aspiration — slope follows opponent mode  
AC-next — as good as our next concealing bid  
**Bait guard** — block deceptive utility spikes

## Evaluation

**Learners** BOA · MAP · MiCRO  
**Stress** time-based NegMAS baselines  
**Sparring** mirror · bait-switch · strong learners  
Strong Concealing vs stress · balanced vs learners & deceptive

## Conclusions

